

Basic Principles of Medicine 1
Module: Foundations to Medicine
Lecture 6

The Action Potential

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Abbreviations

ECF – extracellular fluid

Na-K ATPase – sodium-potassium ATPase (active pump)

mM – millimol (1/1000 mol)

mV – millivolt (1/1000 Volt)

Na_v – voltage-gated sodium channel

E_{ion} – Equilibrium potential / Nernst potential / Diffusion potential

V_m – membrane potential (measured)

RMP – resting membrane potential

log – base 10 logarithm, i.e. log₁₀

Ψ_{ion} – driving force for an ion = V_m - E_{ion}

Ions:

H⁺ – hydrogen

K⁺ – potassium

Na⁺ – sodium

Cl⁻ – chloride

Learning Objectives

- SOM.MK.II.BPM1.1.FTM.3.PHYS.0031 Define the following properties of ion channels: gating, activation, and inactivation.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0032 Contrast the gating of ion selective channels by extracellular ligands, intracellular ligands, stretch, and voltage.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0033 Know the properties of voltage-gated Na^+ , K^+ , and Ca^{2+} channels, and understand that voltage influences their gating, activation, and inactivation.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0034 Understand how the activity of voltage-gated Na^+ , K^+ , and Ca^{2+} channels generates an action potential and the roles of those channels in each phase (depolarization, overshoot, repolarization, hyperpolarization) of the action potential.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0035 Describe ionic basis of an action potential.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0036 Draw a typical action potential and explain why afterhyperpolarization coincides with the greatest fractional K conductance but not the greatest absolute K conductance.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0037 Using the Nernst equation and knowledge of sodium channel voltage dependent gating mechanisms (activation and inactivation gates), contrast the mechanisms leading to weakness and paralysis due to hypo- and hyper-kalemia.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0038 Draw a graph that shows how the voltage sensitivity of the voltage gated sodium channel changes with extracellular calcium concentration.
- SOM.MK.II.BPM1.1.FTM.3.PHYS.0039 Contrast the role that chronic depolarization (e.g. with hyperkalemia) plays in the activation of the K and Na conductances versus rapid depolarization (graded potential towards threshold).

Recommended Reading

Rhoades and Bell, Medical Physiology

- Chapter 3: Section 'Action Potentials'
- the part on fibre myelination will be covered in the NB module in term 2, no need to read now

Outline

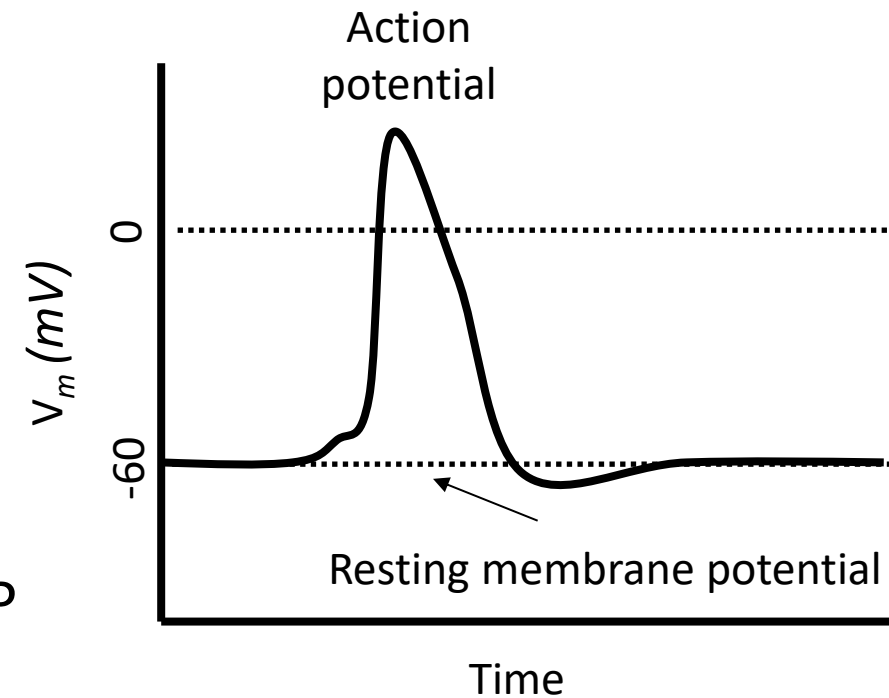
- Overview
- Action Potential (AP) time course and key ion channels
- Channel gating
 - Voltage gated sodium channels
 - Inward rectification and potassium channels
- Phases of the AP
- Clinical correlates of AP
 - Rate of depolarization
 - Potassium (hyper/hypokalemia)
 - Calcium effects on the voltage-gated Na channel
 - Sodium concentrations and effects on AP amplitude

Discussion

- the key permeable ion species at rest is K^+ as it has the largest fractional conductance
- this is why V_m is close to E_K at rest
- hence any substance that targets the resting K^+ conductance (K^+ channels) will have greatest effect on RMP

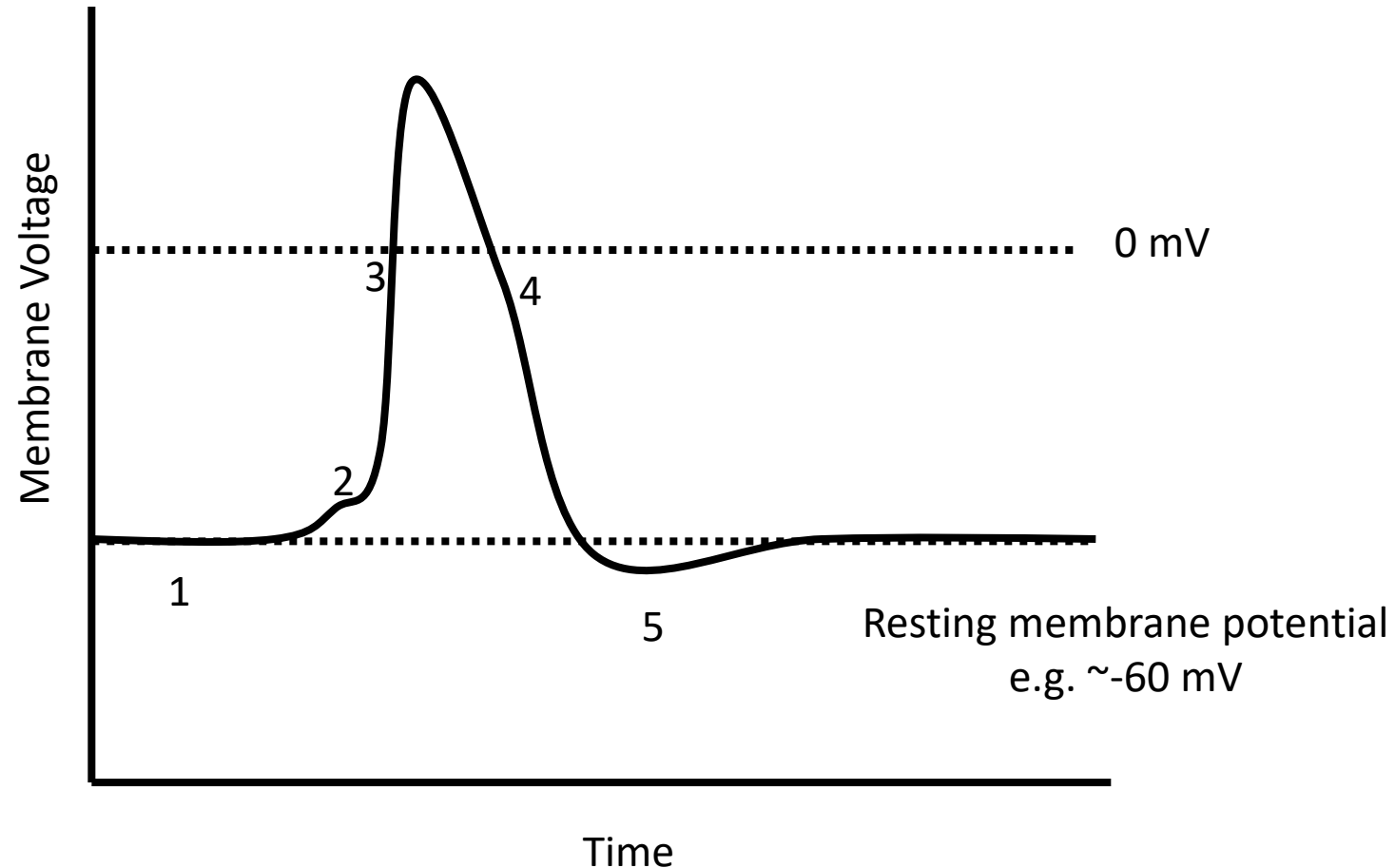
The Action Potential

- a transient, rapid depolarization of the membrane potential
- critical for communication (neurons) and as action trigger (myocytes, endocrine)
- due to swing of V_m from E_K to E_{Na} and back to E_K
 - in cardiac pacemaker cells calcium channels are also important, with Ca^{2+} ion influx playing the same depolarizing role as Na^+ ion influx
- membrane fractional conductances
 - high for K^+ at rest
 - high for Na^+ only during depolarising phase of the AP
 - due to voltage-gated Na^+ and K^+ channels

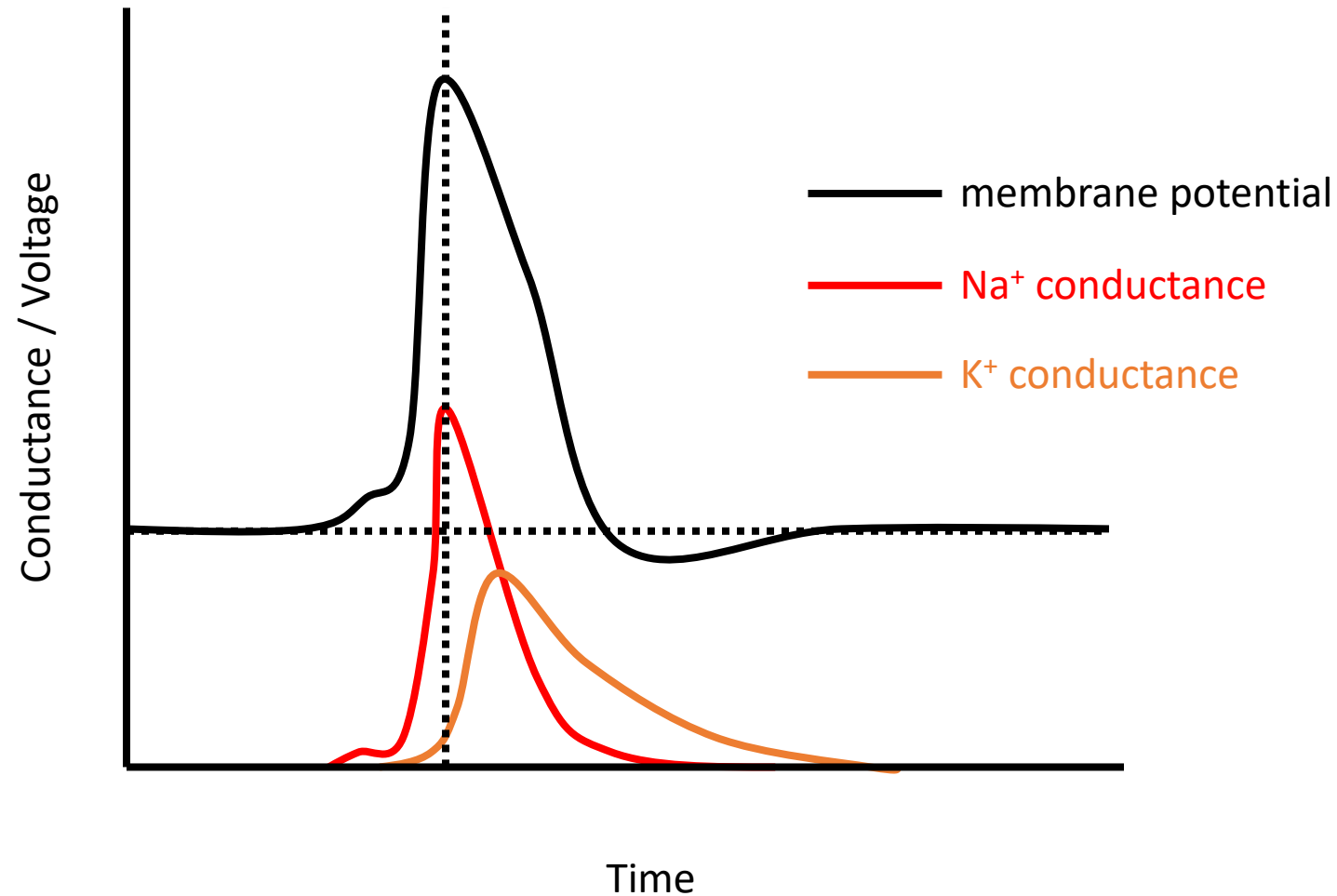


Functional Phases of the AP

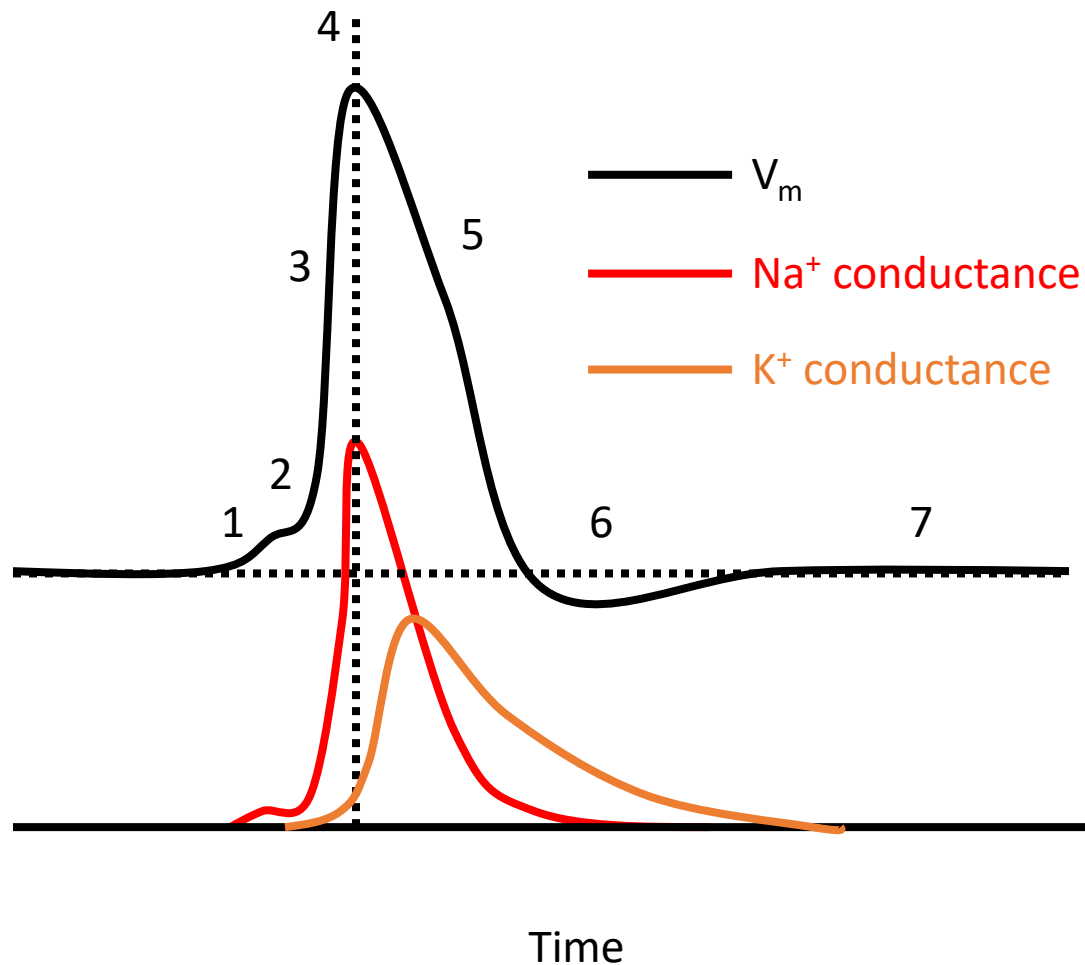
1. Rest membrane potential
2. Graded depolarization phase
3. Rapid depolarization phase
4. Repolarization phase
5. After hyperpolarization phase



Conductances underlying the AP

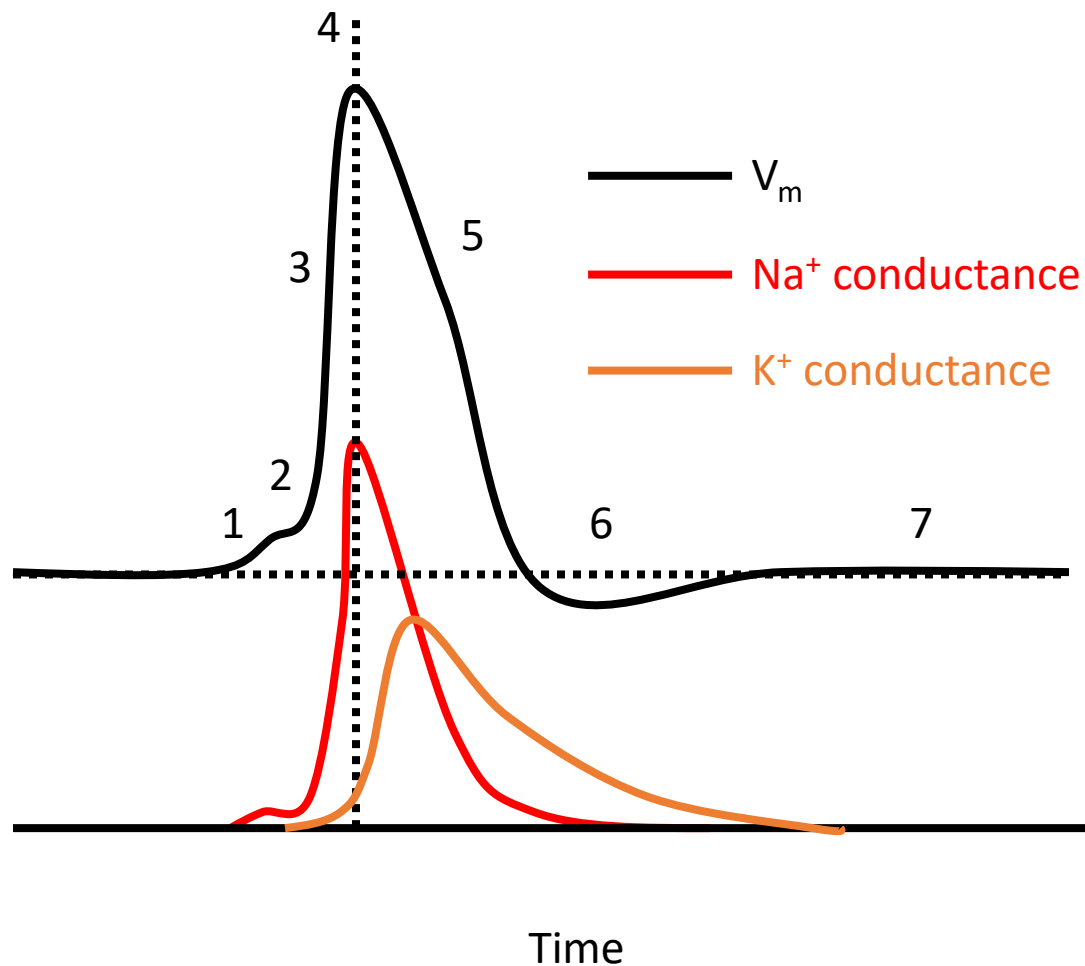


Stages of the Action Potential I



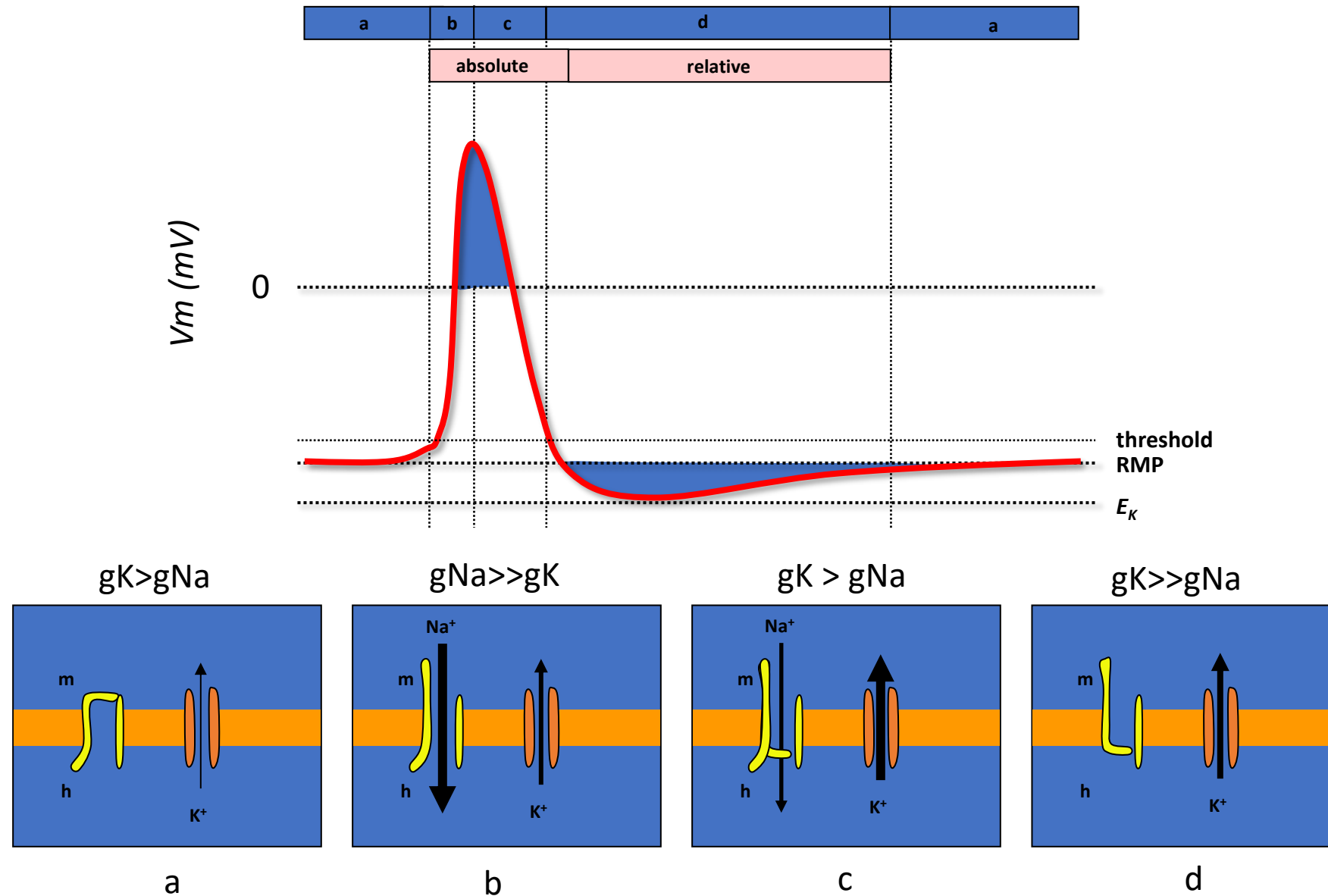
1. Positive charge flows into cell and causes slight depolarization
2. Threshold is reached. Voltage-gated Na^+ channels begin to open fully, leading to more positive charge entering cell and further depolarization.
3. Rapid depolarization as the Na^+ conductance increases dramatically and the membrane potential approaches the equilibrium potential for Na^+ .
4. Peak AP. Na^+ channels are fully open and Na^+ conductance has peaked as V_m approaches E_{Na}

Stages of the Action Potential II



5. Na^+ channels begin to inactivate and Na^+ conductance drops at the same time as the K^+ conductance starts to develop. V_m peaks and starts to swing back towards E_K .
6. Refractory period. Since K^+ channels are open and Na^+ channels are in their inactive state, V_m gets close to E_K , below the resting membrane potential. Due to closed inactivation gates Na^+ channels are blocked, no further APs can be conducted.
7. The channels reset to their normal resting conditions, resting membrane potential is restored. All Na^+ channels are now out of their inactive state and can be reactivated to form another full action potential.

Na⁺ and K⁺ channels during the Action Potential

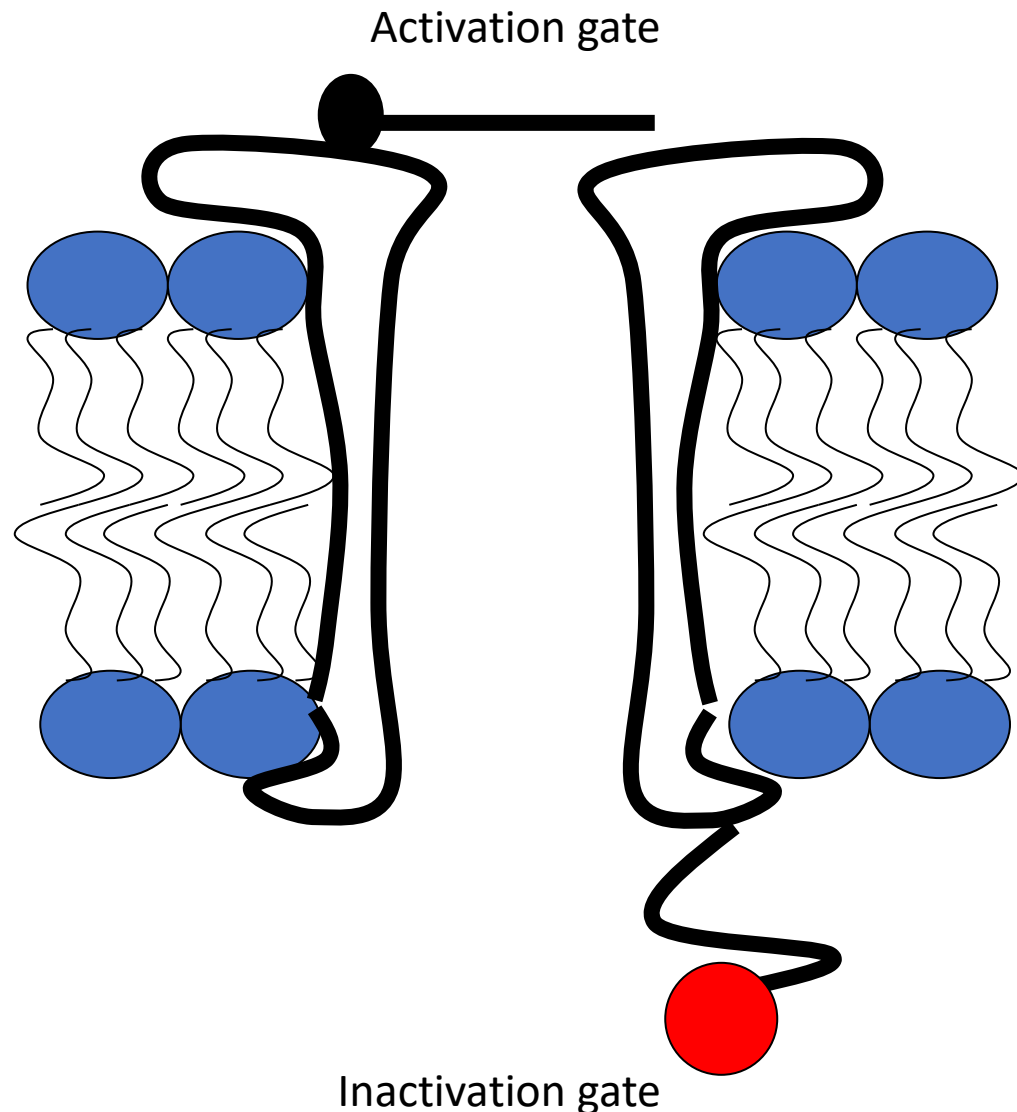


Channel Gating

- opening and closing of the channel (gating) – regulated by electric field, ligands, calcium, phosphatase/kinase activity, cAMP, oxygen, carbon dioxide, etc.
- gates exist in closed or open states, transitions are rapid. There may be more than one gate in a channel
- In some channels flow can occur in either direction (non-rectified), some channels show uni-directional flow (rectified)
- the time a channel spends in the open state determines its open probability (P_o)
- more ion-specific channels in the membrane = higher conductance for that ion
- each channel has a single unit conductance, i.e. a per channel conductance

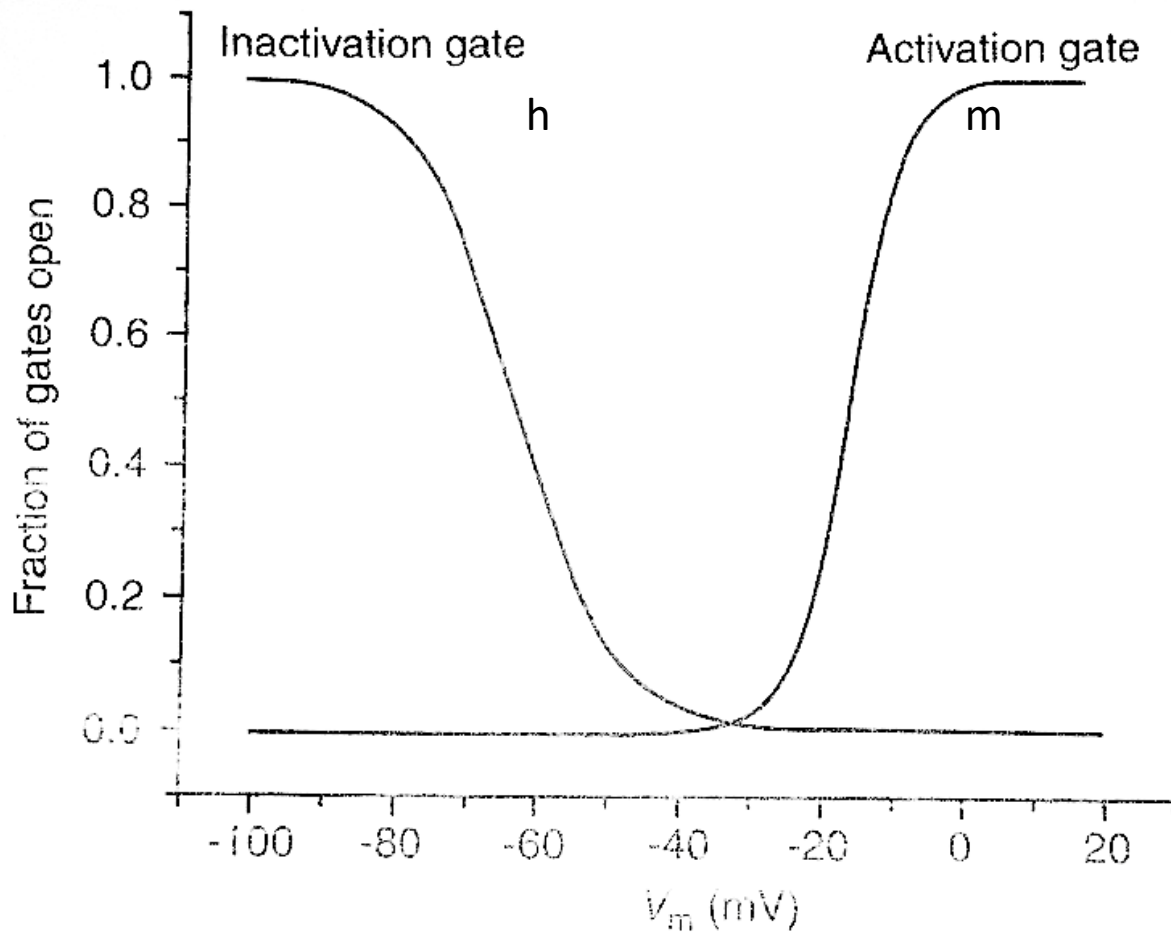
total conductance = number of channels * unit conductance * open probability

The Voltage Gated Na⁺ Channel



- gates are barriers to ion flux
- Na⁺ channel has 2 gates:
 - activation gate (m), extracellular
 - inactivation gate (h), intracellular
 - each gate can block channel independently when closed
- voltage gating:
 - depolarization above threshold ($\sim -50\text{mV}$) opens activation gate
 - inactivation gate closes after slight delay at peak of AP
- the combined action of both gates provides the transient high Na⁺ conductance that carries the AP

Na_v channel gate voltage sensitivity



- Graph depicts the relationship between V_m and open/close states
- No time axis = no kinetics of gate opening in this graph
- Inactivation gate time delay:
 - gates have different rates of operation
 - m gate is quicker than h gate
 - on depolarization the inactivation gates lag behind the activation gate
 - hence both gates are open only for a brief period
- Since both gates are independent, voltage-gated Na channels open only briefly just after above-threshold depolarization

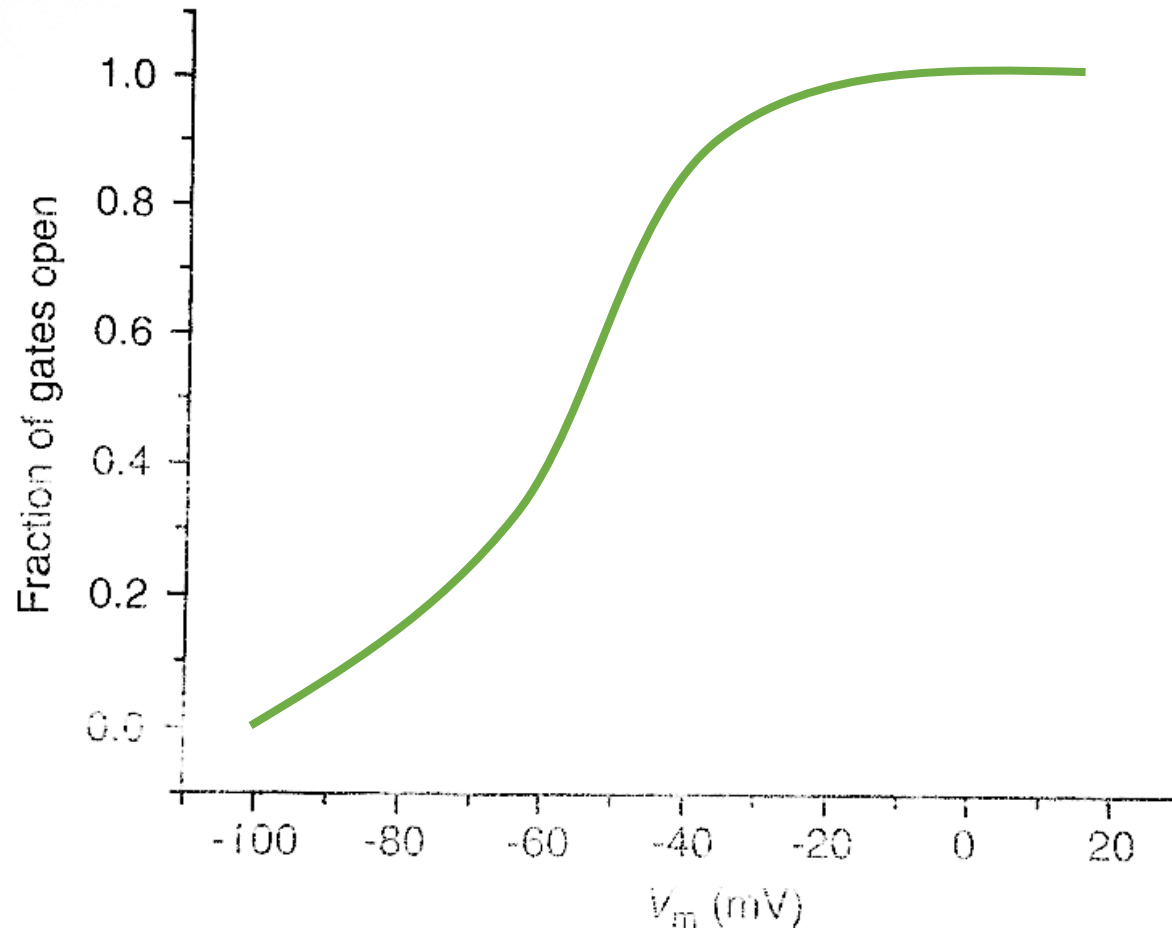
Na_v Gate states during the AP phases

- Resting state
 - activation gate closed and inactivation gate open
 - no flux of Na⁺
- Depolarization phase
 - activation gate open and inactivation gate open
 - channel open, full Na⁺ flux
- Repolarization phase
 - activation gate open but inactivation gate closed
 - no flux of Na⁺
 - channel cannot be activated at this time – absolute refractory period
- Reset to Resting State
 - activation gate closes whilst inactivation gate opens
 - once reset the absolute refractory period is over.

Refractory periods

- Absolute refractory period
 - during repolarization phase when inactivation gates are closed
 - no new AP can be initiated or conducted because Na_v channels are blocked
- Relative refractory period
 - after absolute refractory period during afterhyperpolarization phase
 - starts when the Na_v inactivation gates are open again
 - APs initiation is inhibited due to afterhyperpolarization (increased K^+ conductance)
 - but principally APs can again be initiated since Na_v channels have reset (inactivation gates open)

K⁺ Channel Gates are also Voltage Sensitive

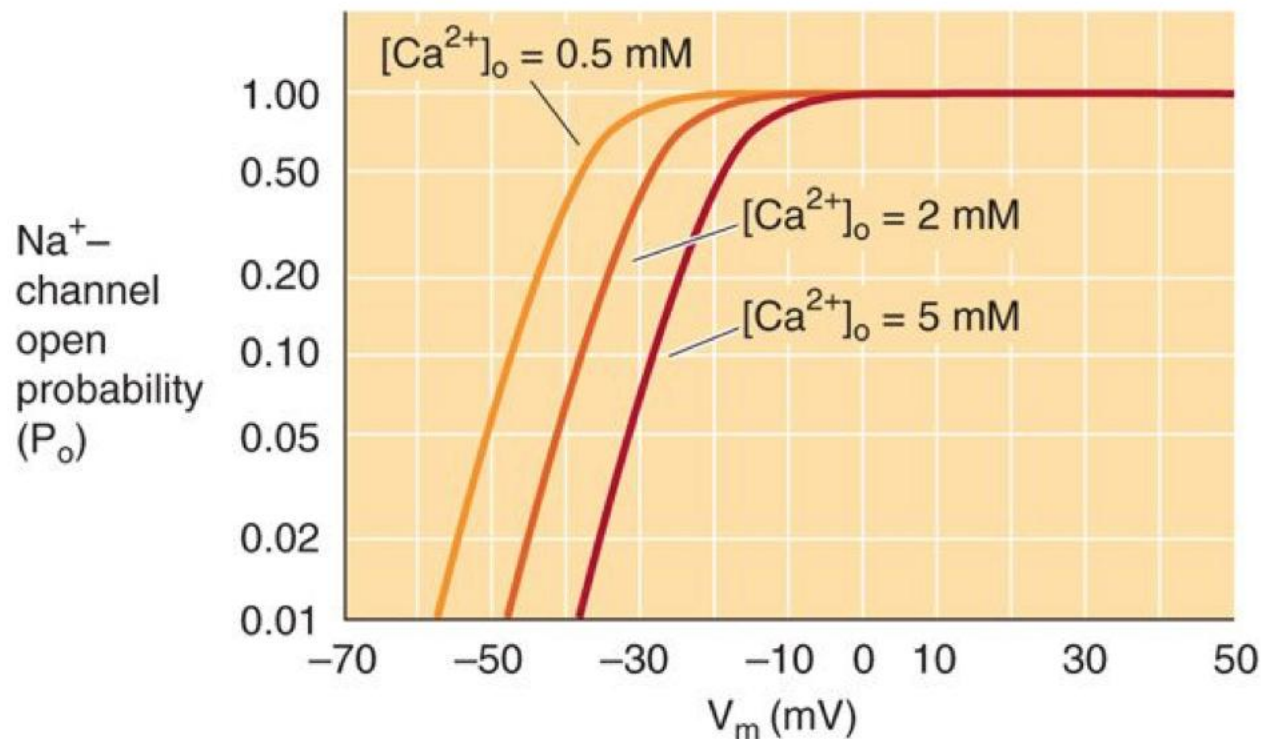


- Voltage sensitive, hence potassium conductance is increased by depolarization
- time and voltage dependent activation of K⁺ channels
- slow time activation compared to Na⁺ channels allows AP to peak before onset of K⁺ current
- delayed rectifying K⁺ channel

Rectification

- delayed rectifier K^+ channels (family of different channels)
- voltage and time dependent activation of K^+ conductance
- highly expressed in neurons
- role is to reset V_m towards E_K by increasing fractional K^+ conductance
- inward rectifying K^+ channels (K_{ir})
- voltage but not time dependent
- current rectification (one-way flow)
- inwards rectifying K^+ channels have strong preference for inward K^+ flux
- expressed in heart and vasculature

Na⁺ channel and Ca²⁺ levels



- Graph shows voltage threshold of activation gate in Na⁺ channel
- sensitive to extracellular Ca²⁺ levels
- Normal ECF [Ca²⁺] ~ 2.1 – 2.8 mmol/L, corresponds with a threshold of around -50 mV
- Hypocalcemia:
 - threshold is lower, closer to RMP → increased excitability. Can lead to tetany (increased muscle tone).
- Hypercalacemia
 - threshold is further from RMP → decreased electrical excitability. May lead to weakness, but uncommon.

Rate of Depolarization is Crucial

- Slow depolarization to threshold leads to
 - sodium channel inactivation gates close in a fraction of the channel population prior to depolarization threshold being reached
 - as a result, a fraction of the sodium channels become 'refractory'
 - delayed rectifier K channels open and increase basal K^+ conductance, stabilizing the membrane potential close to E_K
 - these effects reduce membrane electrical excitability despite the depolarization
- Hence the likelihood of triggering an action potential depends on the rate of depolarization
- Clinically relevant with changes in plasma K^+ concentration

Clinical significance of hyperkalemia

- Normal $[K^+]$ range 3.5 to 5.0 mmol/l
- hyperkalemia means increased extracellular $[K^+]$
- Nernst – depolarization of E_K
- a fraction of Na^+ channels open and close but never reset
 - permanent refractory period, channel inoperable
- K^+ conductance increases and stabilizes membrane potential close to E_K
- Consequences
 - membrane is at a stable, slightly depolarized state with fewer active Na channels
 - action potentials are sluggish or absent
 - weakness and paralysis

$$\begin{aligned} &K^+ \text{ equilibrium potentials} \\ E_K &= -60 \log (120/5) = -82 \\ E_K &= -60 \log (120/10) = -64 \end{aligned}$$

Hypokalemia

$$\begin{aligned} &K^+ \text{ equilibrium potentials} \\ E_K &= -60 \log (120/5) = -82 \\ E_K &= -60 \log (120/2) = -106 \end{aligned}$$

- decreased extracellular $[K^+]$
- Nernst – hyperpolarization of E_K
- consequences at first thought: membrane is at a resting potential that is more negative and thus further away from AP threshold. Depolarizing stimuli should have to be much greater to generate an AP, resulting in lower excitability
- However:
 - AP threshold is also easier to reach because
 - lower RMP $\rightarrow$ less background Na^+ channel opening and closing
 - hence more Na^+ channels are ready to response to depolarizing stimuli
 - More K^+ channels are closed, hence less opposing force to depolarisation
 - steeper slope of depolarization (excitability) and lower threshold potential for full depolarization
 - Na-K ATPase activity also reduced since $[K^+]$ in ECF is reduced
 - the resulting intracellular $[Na^+]$ increase inhibits the efficacy of the Na^+/Ca^{2+} antiporter, the main mechanism to reduce intracellular $[Ca^{2+}]$
 - hence intracellular $[Ca^{2+}]$ remains elevated after action potentials
- Consequences:
 - Muscle weakness, cramps, spasms, respiratory slowing or failure, cardiac arrhythmia, paralysis

Consequences of changes in Sodium

- serum $[\text{Na}^+]$ has little effect on RMP as the fractional sodium conductance at rest is minimal
- but it does change the rate of sodium influx
 - reducing $[\text{Na}^+]$ results in slow influx and smaller, sluggish APs
- Clinically
 - little effect on AP
 - significant effects on cell volume
 - hyponatremia most likely related to kidney function

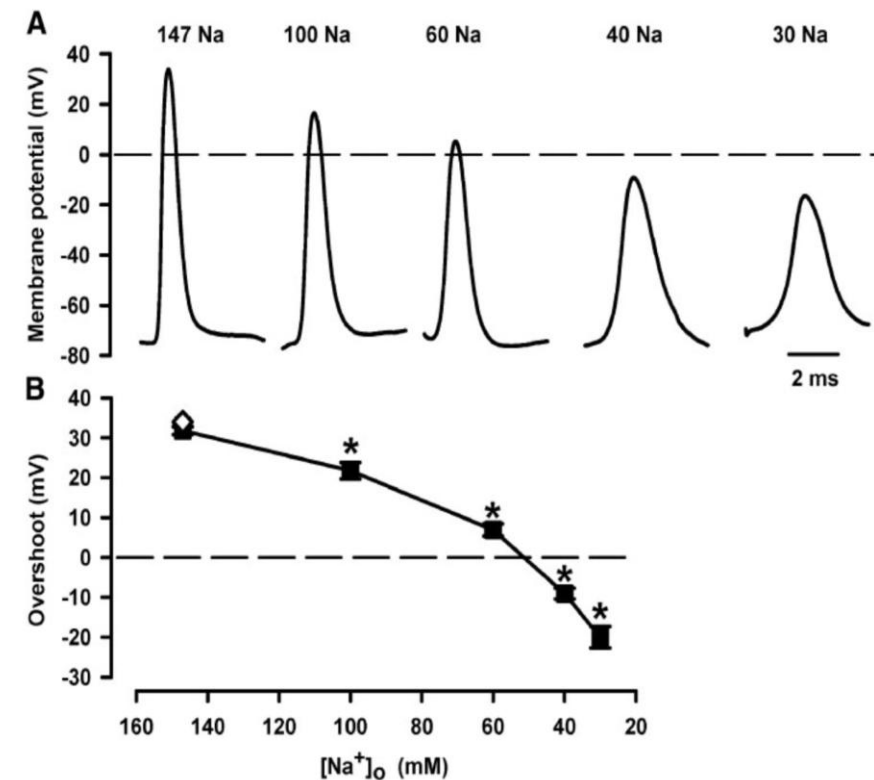


Fig. 6. Effects of lowered $[\text{Na}^+]_o$ on single action potentials in soleus muscle. A: representative traces of action potentials at different $[\text{Na}^+]_o$. B: overshoot- $[\text{Na}^+]_o$ relationship. NMDG was the substitute for Na^+ . Single action potentials were elicited with single 0.3-ms pulses at 10 V (wire electrodes) and measured from fibers located on the surface and in deeper layers of the muscle. ■, overshoot at different $[\text{Na}^+]_o$; ◇, recovery of overshoot after exposure at 30, 40, 60, or 100 mM Na^+ . Horizontal dashed lines represent 0 mV. Values are means $\pm$ SE (numbers of fibers/muscles are given in Table 1). Overshoot was significantly different from that at 147 mM Na^+ ($P < 0.05$, ANOVA and LSD).

Summary

- AP results from transient conductance change towards high fractional conductance for Na^+ before reverting to high K^+ conductance
- Mediated by voltage-gated Na^+ and K^+ channels
- Separate activation and inactivation gates with different kinetics in the voltage-gated Na^+ channel lead to refractory period
 - De-polarization when Na^+ channels open
 - Re-polarization after they close again
 - Refractory period with transient hyper-polarization phase
- Excitable cells can generate APs
 - Neurons
 - Myocytes (cardiac, skeletal, smooth muscle)
 - Some endocrine cells (e.g. pancreatic beta)

small group 3: Self-study element

small group covers membrane and trans-epithelial (TE) potentials
they are not the same, TE potential is across an entire cell

Reading list:

- DLA 4 'Epithelial Function'
- Covering the same concepts as the DLA, read enough of the introduction to understand Figure 1:

Saint-Criq, V., Gray, M.A. Role of CFTR in epithelial physiology. Cell. Mol. Life Sci. 74, 93–115 (2017). <https://doi.org/10.1007/s00018-016-2391-y>

- Read abstract and introduction only, and understand the first figure

Rowe SM, Clancy JP, Wilschanski M. Nasal potential difference measurements to assess CFTR ion channel activity. Methods Mol Biol. 2011;741:69-86. doi: 10.1007/978-1-61779-117-8_6. PMID: 21594779.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3760477/pdf/nihms510689.pdf>

- **Please read these before the small group!**